

SMART: Consolidated Results

Each cell: accuracy / macro-F1 $\pm$ std (%). All numbers are multi-seed (mean over 3 seeds; single-cell and P-NET cohorts alike). The two router columns for cohorts use the Reactome pathway prior as the biological-prior analogue of co-expression.

Dataset	Bio-informed router	General router (no prior)	Adaptive depth (MoR)	Fixed depth	No recursion (K=1)	Vanilla transformer
<i>Genomap single-cell suite</i>						
Tabula Muris	81.1 $\pm$ 1.3 / 69.9 $\pm$ 1.5	81.4 $\pm$ 2.3 / 71.5 $\pm$ 3.0	82.5 $\pm$ 0.3 / 72.7 $\pm$ 1.1	85.3 $\pm$ 1.8 / 77.1 $\pm$ 2.1	82.1 $\pm$ 1.7 / 73.1 $\pm$ 2.4	84.7 $\pm$ 1.3 / 75.5 $\pm$ 0.8
Pancreas	91.9 $\pm$ 1.7 / 59.1 $\pm$ 4.1	90.9 $\pm$ 2.1 / 54.6 $\pm$ 1.1	88.1 $\pm$ 4.0 / 53.5 $\pm$ 1.1	88.9 $\pm$ 6.3 / 53.6 $\pm$ 1.6	90.9 $\pm$ 2.9 / 51.7 $\pm$ 5.2	91.4 $\pm$ 2.2 / 57.3 $\pm$ 1.9
Common	84.1 $\pm$ 1.2 / 70.2 $\pm$ 2.2	81.0 $\pm$ 1.5 / 66.2 $\pm$ 2.9	81.0 $\pm$ 1.5 / 66.2 $\pm$ 2.9	82.9 $\pm$ 0.4 / 68.5 $\pm$ 2.7	81.1 $\pm$ 2.1 / 66.2 $\pm$ 3.7	81.3 $\pm$ 2.1 / 66.6 $\pm$ 4.4
Prototype	94.5 $\pm$ 0.1 / 94.0 $\pm$ 0.2	94.7 $\pm$ 0.3 / 94.2 $\pm$ 0.4	94.7 $\pm$ 0.3 / 94.2 $\pm$ 0.4	94.7 $\pm$ 0.2 / 94.1 $\pm$ 0.3	93.8 $\pm$ 0.2 / 93.2 $\pm$ 0.5	94.7 $\pm$ 0.2 / 94.2 $\pm$ 0.4
Baron	77.8 $\pm$ 7.3 / 57.1 $\pm$ 7.3	80.6 $\pm$ 4.9 / 61.5 $\pm$ 1.8	84.8 $\pm$ 4.8 / 61.1 $\pm$ 1.3	86.4 $\pm$ 2.3 / 66.2 $\pm$ 1.8	81.9 $\pm$ 0.8 / 56.2 $\pm$ 2.6	78.6 $\pm$ 8.6 / 58.8 $\pm$ 4.5
Segerstolpe	17.4 $\pm$ 2.1 / 8.4 $\pm$ 1.4	13.9 $\pm$ 2.5 / 5.7 $\pm$ 0.6	20.0 $\pm$ 6.5 / 7.7 $\pm$ 0.8	17.3 $\pm$ 1.1 / 8.1 $\pm$ 0.7	18.9 $\pm$ 2.2 / 10.1 $\pm$ 0.8	14.9 $\pm$ 0.6 / 6.7 $\pm$ 1.6
Lung	79.6 $\pm$ 0.5 / 72.6 $\pm$ 1.6	78.3 $\pm$ 2.0 / 70.7 $\pm$ 4.2	78.3 $\pm$ 2.0 / 70.7 $\pm$ 4.2	77.6 $\pm$ 1.6 / 70.3 $\pm$ 1.9	78.5 $\pm$ 1.8 / 70.3 $\pm$ 1.3	79.8 $\pm$ 0.2 / 72.5 $\pm$ 2.4
Oesophagus	80.7 $\pm$ 2.0 / 56.5 $\pm$ 5.0	80.6 $\pm$ 1.4 / 56.1 $\pm$ 3.8	80.6 $\pm$ 1.4 / 56.1 $\pm$ 3.8	81.0 $\pm$ 0.2 / 55.1 $\pm$ 1.3	79.3 $\pm$ 1.8 / 56.5 $\pm$ 4.0	81.3 $\pm$ 0.7 / 55.8 $\pm$ 2.9
Spleen	56.3 $\pm$ 3.0 / 48.5 $\pm$ 2.4	58.9 $\pm$ 1.1 / 50.4 $\pm$ 2.3	58.9 $\pm$ 1.1 / 50.4 $\pm$ 2.3	53.2 $\pm$ 2.7 / 46.7 $\pm$ 3.5	56.0 $\pm$ 1.1 / 48.8 $\pm$ 2.7	56.6 $\pm$ 2.4 / 49.4 $\pm$ 3.6
T-cell	65.4 $\pm$ 2.5 / 48.7 $\pm$ 1.4	69.2 $\pm$ 2.3 / 50.0 $\pm$ 3.2	67.6 $\pm$ 1.5 / 49.7 $\pm$ 2.0	65.2 $\pm$ 1.4 / 49.3 $\pm$ 1.8	67.3 $\pm$ 1.7 / 50.4 $\pm$ 1.9	65.8 $\pm$ 4.6 / 47.7 $\pm$ 2.4
<i>P-NET / Reactome multi-omics cohorts</i>						
Prostate	66.0 $\pm$ 4.5 / 55.0 $\pm$ 9.0	68.0 $\pm$ 3.1 / 54.3 $\pm$ 9.3	68.6 $\pm$ 2.4 / 55.5 $\pm$ 8.7	62.2 $\pm$ 3.0 / 58.3 $\pm$ 2.4	61.6 $\pm$ 2.4 / 50.4 $\pm$ 5.9	67.8 $\pm$ 5.7 / 56.8 $\pm$ 12.3
BLCA	59.7 $\pm$ 7.3 / 41.6 $\pm$ 3.2	59.3 $\pm$ 5.6 / 41.8 $\pm$ 4.5	58.8 $\pm$ 7.6 / 41.1 $\pm$ 2.9	62.6 $\pm$ 3.1 / 41.3 $\pm$ 2.9	55.6 $\pm$ 16.6 / 36.0 $\pm$ 8.4	66.7 $\pm$ 1.0 / 40.0 $\pm$ 0.4
STAD	46.6 $\pm$ 8.8 / 45.0 $\pm$ 7.5	50.6 $\pm$ 6.0 / 45.4 $\pm$ 7.3	48.2 $\pm$ 8.1 / 44.7 $\pm$ 7.7	53.8 $\pm$ 5.4 / 49.4 $\pm$ 3.5	45.8 $\pm$ 2.0 / 40.7 $\pm$ 2.0	51.8 $\pm$ 4.3 / 47.8 $\pm$ 3.5
PanCan	62.0 $\pm$ 4.9 / 57.9 $\pm$ 5.9	62.1 $\pm$ 6.9 / 57.6 $\pm$ 6.9	63.0 $\pm$ 4.3 / 58.4 $\pm$ 4.0	73.0 $\pm$ 2.5 / 69.4 $\pm$ 2.3	64.2 $\pm$ 2.0 / 59.9 $\pm$ 1.9	56.3 $\pm$ 6.5 / 52.1 $\pm$ 8.7

Statistical significance analysis.

- Biological prior (co-expression vs degree-matched random-graph control, paired over all dataset $\times$ seed): mean Δ F1 = +0.28 pts, 95% CI [-1.2, +1.7], p = 0.871, Cohen's d_z = +0.07; 0/10 datasets significant after Holm-Bonferroni. Verdict: NOT significant – statistically indistinguishable from a random graph.
- Adaptive depth: recursion vs single pass (K=1) Δ F1 = +0.60 (p = 0.337); adaptive vs fixed depth is not formally equivalent at a 1.0-pt margin (TOST p = 0.278); mean compute saving 31% (p = 0.000 that saving > 0). Verdict: the compute reduction is the decisive, significant effect; accuracy/depth gains are modest.